

Climate Volatility, Agricultural Pathway, and Modern Demographic Outcomes

A Cross-Country Investigation Across Pre-Industrial and Modern Windows

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Abstract

A country-level panel is assembled of five pre-industrial channels of long-run development: a four-element climate bundle ($\bar{T}$, $\bar{P}$, σ_v^T , σ_v^P over 1421–1750 from the ModE-RA paleo-reanalysis); an eight-locus functional-allele bundle (LCT, ADH1B, AMY1, EDAR, DARC, SLC24A5, HBB, FADS1/2) indexing specific Holocene-era selective adaptations; a Neolithic-ancestry fraction indexing the share of modern ancestry derived from any agricultural-Neolithic source population (Lazaridis, Patin, Reich, Yang, Skoglund et al.); ancestral crop-yield potential (Galor and Özak, 2016); and pre-1500 pandemic intensity. The two modern population-genetics substrates replace the Ashraf-Galor (2013) ancestry-adjusted predicted-heterozygosity measure as headline substrates, addressing a sustained critique of the AG framework that the scalar-diversity interpretation reified abstract genetic variation as a deep determinant of development. The panel is used to investigate how each substrate shapes a six-outcome vector spanning the Malthusian-to-post-Malthusian transition: log population density in 1500 CE (Malthusian-era proxy), log population density in 2025 (cumulative-productivity proxy), log population growth 1950–2025, urbanisation-share change 1950–2025, log GDP per capita 2015, and demographic-transition timing. Five Westfall-Young FWER-corrected cells survive at $p < 0.05$ across the 30-cell joint- F mediation matrix. Ancestral crop yield is the dominant predictor of modern population growth ($F = 27.0$, $p_{WY} < 0.001$) and of population density at 1500 ($F = 15.5$, $p_{WY} = 0.002$) and 2025 ($F = 10.4$, $p_{WY} = 0.030$), spanning the full demographic-density trajectory. Neolithic-ancestry fraction is the dominant new finding—a strong predictor of log population density 2025 ($F = 17.9$, $p_{WY} < 0.001$) and demographic-transition timing ($F = 13.2$, $p_{WY} = 0.005$). The functional-allele bundle contributes the largest Shapley R^2 on log GDP per capita (0.175) and demographic-transition timing (0.122)—substantially larger than the predicted-heterozygosity contribution it replaces—but its joint significance is spread across the eight collinear loci. Within the bundle, FADS1/2, AMY1, and DARC are the largest individual contributors to log GDP per capita. The climate bundle dominates Shapley R^2 on log population density 1500 (0.088) and urbanisation change (0.076), and its mediation share on density 1500 is 0.75—the cleanest illustration that the climate channel operates specifically through agricultural-pathway selection at the Malthusian density margin. The five FWER-surviving findings are robust to climate-window placebos, cumulative conflict-exposure and volcanic-radiative-forcing controls, and a climate-A partialling that breaks the GAEZ-implied correlation between ancestral crop yield and the climate bundle. The investigation places four pre-existing causal-identification literatures on a single panel and reveals a clean temporal structure: climate at the Malthusian density margin via pathway mediation; ancestral crop yield across the whole density trajectory; Neolithic-ancestry fraction at the cumulative-density and demographic-transition margins; and

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the functional-allele bundle as the modern, defensible reinterpretation of what the Ashraf-Galor measure was capturing.

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1 Introduction

A country’s pre-industrial conditions correlate strongly with its modern demographic outcomes. The empirical literature has identified at least four distinct channels through which this correlation operates. Ashraf and Galor (2013) argue that the ancestral genetic diversity of a country’s population, predicted by the out-of-Africa migration history of its constituent ancestral populations and ancestry-adjusted to year-2000 composition (Putterman and Weil, 2010), drives an inverted-U relationship with development. Galor and Özak (2016) argue that the ancestral suitability of a country’s land for pre-1500 cultivated crops, measured from Global Agro-Ecological Zones (GAEZ) projections under pre-Columbian climate, shapes long-run economic outcomes through its effect on time-preference selection. The Voigtländer-Voth post-Black-Death literature (Voigtländer and Voth, 2013) argues that pre-1500 pandemic exposure had persistent effects on factor ratios and subsequent demographic-economic trajectories. The companion paper (Alonso Ortiz and Da-Rocha, 2026)—hereafter LS—argues that pre-industrial inter-annual climate volatility 1421–1750 selected agricultural pathways and through them shaped both pre-industrial Malthusian dynamics and the modern demographic record. We treat the climate channel as a four-element bundle—mean temperature $\bar{T}$, mean precipitation $\bar{P}$, inter-annual temperature volatility σ_v^T , and inter-annual precipitation volatility σ_v^P , all over 1421–1750—rather than reducing it to σ_v^T alone, both to put the climate channel on equal footing with the other three substrates (each of which is itself an index summarising multiple primitives) and because the Matranga (2024) storage mechanism that motivates the climate channel concerns the full bundle.

Each of these literatures has its own causal-identification strategy and its own preferred outcome. Their relationship to one another, on a single panel with internally consistent measurement, has not been investigated systematically. Several questions follow naturally. Does the climate bundle’s predictive contribution survive once ancestry-derived substrates and ancestral crop-yield potential are conditioned in the same regression? Does the agricultural pathway typology, which the LS companion paper estimates is selected by climate volatility, mediate the climate bundle’s effect on long-run outcomes? Do the substrates matter at different points in the Malthusian-to-post-Malthusian transition—some for pre-industrial population density, others for post-industrial per-capita income? Are the modern population-genetics substrates (functional-allele bundle, Neolithic-ancestry fraction) more informative on the panel than the scalar predicted-heterozygosity measure they replace?

This paper investigates these questions on a 196-country panel that joins the climate bundle with the two modern population-genetics substrates, ancestral crop-yield potential, and pre-1500 pandemic intensity. The six outcomes span the demographic transition: log population density in 1500 CE (Malthusian-era proxy), log population density in 2025 (cumulative-productivity proxy), log population growth 1950–2025, urbanisation-share change 1950–2025, log GDP per capita 2015, and demographic-transition timing. The first two outcomes are drawn from HYDE 3.5 (Goldewijk et al., 2017); the latter four from UN WPP, WUP, Maddison, and Gapminder/OWID sources. The K=5 agricultural-pathway typology of LS enters as the candidate mediator. We frame the exercise as an *investigation* rather than a horserace with a designated winner: each substrate has its own

causal-identification literature and we borrow their assumptions rather than re-litigate them. The contribution is to put them side-by-side and ask which carries unique-variance weight on which outcomes.

Five findings survive Westfall-Young family-wise error-rate correction across the 30 substrate-by-outcome cells of the mediation matrix:

First, ancestral crop-yield potential is the dominant predictor of modern population growth, and a robust predictor of population density at both 1500 and 2025—spanning the full demographic-density trajectory. On log population growth 1950–2025 the FWER-corrected joint F -statistic on the mediated coefficient is 27.0 ($p_{WY} < 0.001$), the largest single cell in the matrix. On log population density 1500 the Shapley R^2 is 0.012 ($F = 15.5$, $p_{WY} = 0.002$); on log population density 2025 it is essentially zero ($F = 10.4$, $p_{WY} = 0.030$) but the joint coefficient retains FWER significance. Together these findings extend the Galor-Özak (2016) result—originally framed through the time-preference channel—to the cross-country demographic-density trajectory: pre-1500 caloric potential of cultivable crops predicts population density at the Malthusian steady state, at the post-industrial steady state, and the convergence path that connects them.

Second, Neolithic-ancestry fraction is the dominant predictor of cumulative population density and demographic-transition timing. A scalar measure of the fraction of modern ancestry derived from agricultural-Neolithic source populations—aggregated globally from Lazaridis (Eurasia), Patin and Lipson (Africa), Reich and Posth (Americas), Yang and Wang (East Asia), and Skoglund and Bergström (Pacific)—is a strong predictor of log population density 2025 ($F = 17.9$, $p_{WY} < 0.001$) and demographic-transition timing ($F = 13.2$, $p_{WY} = 0.005$). Substantively, populations whose ancestors underwent an agricultural-Neolithic transition (Anatolian Neolithic farmer, Bantu, Yangtze rice-farmer, Mesoamerican maize-farmer, Andean potato-farmer, Austronesian) reached higher cumulative density and completed the demographic transition earlier than populations descended primarily from pre-Neolithic forager lineages (WHG, EHG, Khoisan, Papuan, Aboriginal Australian). This finding is the natural global extension of the within-Eurasia §6 ancestry-decomposition exercise, and is substantively more interpretable than the predicted-heterozygosity measure it replaces.

Third, the functional-allele bundle contributes the largest Shapley R^2 on log GDP per capita and demographic-transition timing, providing a functionally-interpretable alternative to the Ashraf-Galor heterozygosity scalar. An eight-locus bundle of derived-allele frequencies for SNPs with established functional and economic-history channels—LCT (lactase persistence), ADH1B (East Asian fast alcohol metabolism), AMY1 (starch digestion), EDAR (East Asian morphology), DARC (Duffy-null malaria resistance), SLC24A5 (skin pigmentation/latitude), HBB (sickle-cell malaria resistance), and FADS1/2 (post-agricultural diet adaptation)—contributes Shapley R^2 0.175 on log GDPpc and 0.122 on demographic-transition timing, both substantially larger than the predicted-heterozygosity contribution they replace. Within the bundle, FADS1/2 (0.041) and DARC (0.037) are the largest individual contributors to log GDPpc, while FADS1/2 and LCT dominate demographic-transition timing. The bundle’s joint significance is split across the eight collinear loci, so the bundle does not survive FWER correction as a single F -test even though its Shapley contribution is large. We report a malaria-loci-removed sensitivity (Section 6.5) to isolate the agricultural-adaptation signal from the tropical-disease signal.

Fourth, the climate bundle dominates Shapley R^2 at the Malthusian density margin and operates through pathway selection. The bundle’s Shapley R^2 is 0.088 on log population density 1500 (the largest of the five substrates) and 0.076 on urbanisation change. The mediation share on density 1500 is 0.75 and on population growth is 0.92, the cleanest illustration in the matrix that the climate channel operates specifically through agricultural-pathway selection rather than as a direct channel. Within the bundle (Table 5), σ_v^T specifically drives the 1500-density contribution, consistent with the Matranga (2024) storage-and-intensification mechanism.

Fifth, the H_pred coefficient evolution test confirms that the new modern-genetics substrates absorb what the AG measure was capturing. On the full 162-country panel, when the functional-allele bundle is added to a regression of log GDPpc on H_pred and controls, the H_pred coefficient becomes unstable and changes sign while R^2 jumps from 0.46 to 0.62—a pattern consistent with the functional alleles being highly collinear with what H_pred measures, but expressing it as a vector of interpretable selection signatures rather than a scalar diversity index.

Pathway mediation is well-behaved for the climate-bundle and ancestral-yield cells but exhibits suppressor structure for the other substrates in baseline pooled specifications; continent fixed effects collapse the suppressor cells. The five FWER-surviving findings are robust to a battery of additional checks: climate-window placebos, the addition of cumulative conflict and volcanic-radiative-forcing controls, R1b-M269 frequency, Lazaridis ancient-DNA-derived ancestry components, and a climate-A partialling robustness that breaks the GAEZ-implied correlation between ancestral crop yield and the climate bundle.

Contributions. Three contributions accompany the substantive findings. *Methodologically*, we build a country-level deep-determinants panel that puts five major pre-industrial channels on internally consistent measurement, treats climate and functional alleles as coalition players in a grouped-Shapley decomposition, and uses Westfall-Young multi-testing correction across a 30-cell matrix spanning the Malthusian-to-post-Malthusian transition. *Substantively*, we replace the Ashraf-Galor predicted-heterozygosity measure with two functionally-interpretable alternatives drawn from modern ancient-DNA evidence—the functional-allele bundle and Neolithic-ancestry fraction—and identify ancestral crop yield as the dominant predictor across the whole demographic-density trajectory. *Conceptually*, we demonstrate a clean temporal-moment structure: climate at the Malthusian density margin via pathway mediation; ancestral crop yield across the whole density trajectory; Neolithic-ancestry fraction at the post-industrial cumulative-density and demographic-transition margins; and the functional-allele bundle as the modern reinterpretation of what the AG diversity measure was capturing.

The paper proceeds as follows. Section 2 sketches the climate-bundle/pathway-selection skeleton that motivates the five-substrate investigation. Section 3 describes the master horserace panel and its construction, including the climate bundle, the functional-allele bundle, the Neolithic-ancestry fraction, and the two population-density outcomes. Section 4 reports the Shapley-Owen variance decomposition (Exercise 1), including a within-bundle sub-decomposition across the four climate primitives. Section 5 runs the pathway-mediation analysis with and without continent fixed effects (Exercise 2). Section 6 reports the robustness battery: sub-sample stability, pathway-cluster source, climate-window placebos, continent fixed effects, leave-one-out, Westfall-Young FWER, climate-A partialling, cumulative conflict and volcanic exposure controls, and direct ancient-DNA-derived ancestry-composition controls. Section 7 discusses implications for the deep-determinants literature; Section 8 concludes.

2 Climate, pathway selection, and the demographic transition: a UGT skeleton

Consider a country i with a pre-industrial climate endowment $\mathbf{C}_i = (\bar{T}_i, \bar{P}_i, \sigma_{v,i}^T, \sigma_{v,i}^P)$ summarising the central tendency and inter-annual variability of temperature and precipitation over a multi-century pre-industrial window. Climate shapes agricultural-system selection: societies facing high volatility or strong seasonality face strong incentives to adopt storage-intensive crop systems with granaries and reserves, while societies in low-volatility benign climates can rely on shorter-horizon production and societies in arid or marginally cultivable climates default to pastoralism (Matranga,

2024). The climate bundle therefore maps into a discrete agricultural pathway $\tau_i \in \{\text{intensive crop, pastoral, mixed late, early extensifier}\}$ via

$$\tau_i = \tau^*(\mathbf{C}_i, \mathbf{X}_i; \theta) + \xi_i \quad (1)$$

where $\mathbf{X}_i$ is a vector of geography controls (absolute latitude, log land area, ruggedness, log Neolithic distance, landlocked indicator) and ξ_i captures historical contingency. The pathway τ_i in turn shapes pre-industrial demographic dynamics through a Malthusian regression in the spirit of Ashraf and Galor (2011):

$$\Delta \ln P_{i,t} = \alpha_i + \beta(\tau_i) d_{i,t-1} + \gamma(\tau_i) T_{i,t} + \delta(\tau_i) \mathbf{C}_i + \eta(\tau_i) h_{i,t} + \varepsilon_{i,t}, \quad (2)$$

with pathway-specific coefficients $\beta(\tau), \gamma(\tau), \delta(\tau)$, and a human-capital term $\eta(\tau) h_{i,t}$ that captures the rate at which the Malthusian feedback weakens with secular accumulation. The companion paper (Alonso Ortiz and Da-Rocha, 2026) estimates equation (2) directly on the 1421–1950 panel; we treat it as motivational here and ask what its long-run implications for both pre-industrial and modern outcomes look like once competing channels are conditioned.

Outcomes $y_i^{(k)}$ span the Malthusian-to-post-Malthusian transition: *Malthusian-era* (log population density at 1500, where productivity translates into people rather than per-capita output), *cumulative* (log population density at 2025, reflecting productivity accumulated through industrialisation), and four post-industrial outcomes (population growth 1950–2025, urbanisation change, log GDP per capita 2015, demographic-transition timing). They inherit two types of climate dependence: *indirect* through τ_i via the pathway-specific coefficients of (2), and *direct* through any persistent channel that bypasses pathway. We investigate three competing pre-industrial channels alongside the climate bundle:

- **Predicted heterozygosity** H_i (Ashraf and Galor, 2013), the ancestry-adjusted predicted population genetic diversity of a country’s modern population, which enters as a direct channel on modern outcomes not mediated by agricultural pathway.
- **Ancestral crop yield** A_i (Galor and Özak, 2016), the gridded pre-1500 caloric-yield potential of Old-World crops aggregated to country level, a competing pre-industrial channel that may dominate the climate bundle on agricultural-system outcomes.
- **Pre-1500 pandemic intensity** Π_i , a Voigtländer-Voth-style historical mortality-pressure measure capturing exposure to the Antonine, Cyprian, Justinianic, and Black-Death pandemics with their post-1500 recurrences.

The empirical exercises decompose cross-country variance in $y_i^{(k)}$ across the five substrates (with climate and functional alleles as coalition players) and ask which survive multi-testing correction.

The model delivers three predictions that structure the empirical work. **Prediction 1.** The Shapley R^2 contribution of the climate bundle $\mathbf{C}_i$ is positive and largest on outcomes most closely tied to the Malthusian steady state—population density at 1500 above all—because the bundle’s influence operates through agricultural-system selection that determines pre-industrial carrying capacity. **Prediction 2.** Adding the $K=5$ pathway-dummy vector $\mathbf{P}_i$ shrinks the bundle’s marginal contribution sharply on outcomes where the mechanism is mediated, and the mediation share approaches one for density 1500. The shrinkage on income outcomes is smaller because pathway only partially mediates the bundle’s effect on post-industrial productivity. **Prediction 3.** Within the climate bundle, inter-annual temperature volatility σ_v^T is the dominant primitive for Malthusian-era density (since storage demand and intensification respond to volatility), while mean precipitation $\bar{P}$ is the dominant primitive for modern population-growth and urbanisation outcomes (since these reflect water availability for modern agricultural and urban systems).

3 Data

The paper rests on a country-level panel of five pre-industrial substrates, six outcomes spanning the demographic transition, and a standard battery of geography controls, covering 196 countries (162 with non-missing values on all five substrates).

3.1 Channels

Climate bundle $\mathbf{C}_i = (\bar{T}_i, \bar{P}_i, \sigma_{v,i}^T, \sigma_{v,i}^P)$ is the four-element vector of country-level pre-industrial climate primitives over 1421–1750. Mean temperature $\bar{T}_i$ and mean annual precipitation $\bar{P}_i$ are absolute country-level averages computed from the monthly ModE-RA paleo-reanalysis (Franke et al., 2023) aggregated to country level by cosine-latitude-weighted area sums; the CRU TS 4.09 (Harris et al., 2020) 1901–1950 country-month climatology is added back to ModE-RA anomalies to yield absolute T and P values. Inter-annual volatility components $\sigma_{v,i}^T$ and $\sigma_{v,i}^P$ are the country-level standard deviations of annual mean temperature and annual mean precipitation over the 1421–1750 window. The country-level monthly panel is constructed and validated in the companion paper (Alonso Ortiz and Da-Rocha, 2026): median country-month temperature correlation between ModE-RA and ERA5 in their 1950–2008 overlap is 0.971, with 77.6% of countries exceeding $r = 0.90$.

The within-bundle correlation structure is moderate (Table 2): σ_v^T correlates -0.67 with $\bar{T}$ (cold places are more volatile), $\bar{P}$ correlates $+0.66$ with σ_v^P (wetter places have more precipitation variability), and all six pairwise correlations sit in $|r| \in [0.24, 0.67]$. This justifies treating the four primitives as a coalition player rather than expanding them into separate substrates: they share substantial common variation, and the storage-and-intensification mechanism that motivates the climate channel concerns the bundle jointly.

Functional-allele bundle. An eight-element vector of country-level derived-allele frequencies at SNPs with established functional and economic-history channels: LCT/MCM6 rs4988235 (lactase persistence, dairy-pastoral package); ADH1B rs1229984 (His47, East Asian fast alcohol metabolism); AMY1 (salivary amylase copy-number proxy, starch digestion); EDAR rs3827760 (V370A, East Asian morphology); DARC rs2814778 (Duffy-null, malaria resistance); SLC24A5 rs1426654 (Ala111Thr, skin pigmentation / latitude); HBB rs334 (HbS sickle, malaria resistance); and FADS1/2 rs174570 (long-chain PUFA conversion, post-agricultural diet adaptation). Per-locus frequencies are assembled from the primary population-genetics literature (Itan et al. 2010, Liebert et al. 2017, Howes et al. 2011, Piel et al. 2010, Norton et al. 2007, Sabeti et al. 2007, Mathieson et al. 2015, Peng et al. 2010, Perry et al. 2007) for 19 anchor populations spanning global ancestry; country-level frequencies are computed by Putterman-Weil (2010) ancestry-weighted summation, identical in construction logic to the Ashraf-Galor pdiv_aa measure but using a vector of functional alleles in place of the scalar predicted diversity. Treated as a Shapley coalition player parallel to the climate bundle. The bundle replaces the Ashraf-Galor predicted-heterozygosity measure as a headline substrate, addressing the sustained critique of the AG framework that the scalar-diversity interpretation reified abstract genetic variation as a deep determinant of development.

Neolithic-ancestry fraction ν_i is the country-level share of modern ancestry derived from populations that underwent an agricultural-Neolithic transition of any kind, versus pre-Neolithic forager ancestry. The post-Neolithic share aggregates Anatolian Neolithic farmer, Iranian Neolithic, Caucasus-Neolithic, and Yamnaya steppe-pastoralist ancestry in Eurasia (Lazaridis et al., 2014); Bantu agriculturalist ancestry in Sub-Saharan Africa (Patin et al., 2017; Lipson et al., 2020); Mesoamerican maize-farmer and Andean potato-farmer ancestry in the Americas (Reich et al., 2012; Posth et al., 2018); Yangtze rice-farmer and Han-Neolithic ancestry in East Asia (Wang et al. 2019; Yang et al. 2020); and Austronesian ancestry in the Pacific (Skoglund et al., 2016; Bergström et al.,

2017). Pre-Neolithic lineages excluded from the fraction include Western Hunter-Gatherer (WHG), Eastern Hunter-Gatherer (EHG), Khoisan, Papuan, and Aboriginal Australian. For 86 of 186 countries the fraction is computed directly from cited regional decompositions; for the remaining 100 countries it is imputed from the regional modal classification, with the imputation flag documented per-country in the build script. A strict-agriculturalist variant excludes Yamnaya/steppe ancestry from the fraction; both variants are reported in robustness.

Predicted heterozygosity H_i remains in the panel as a baseline comparison but is not a headline substrate in horserace v3. It is the ancestry-adjusted predicted heterozygosity of Ashraf and Galor (2013), drawn directly from their AER replication archive. We demote H_i from headline substrate to the coefficient-evolution robustness exercise (Section 6.4), where we document how its coefficient changes as the modern-genetics substrates are progressively added.

Ancestral crop yield A_i is the country-level mean of the pre-1500 Caloric Suitability Index (Galor and Özak, 2016), downloaded from Zenodo (ozak/Caloric-Suitability-Index, deposit 14714917) and using the `pre15002AverageCalories0mean` variable. We work with the log-transformed measure $\ln(A_i + 1)$ to handle countries with zero usable agricultural cells (Iceland, the Gulf states). The Caloric Suitability Index is itself constructed from GAEZ projections under pre-Columbian climate; mean temperature and mean precipitation therefore enter both the climate bundle and (implicitly) the A construction. We pre-emptively flag this in the robustness battery (Section 6.6), where we report a climate- A partialling robustness column that regresses A on $(\bar{T}, \bar{P})$ and uses the residual as a “climate-residualised” ancestral yield. The five FWER-surviving findings are unchanged under this robustness specification.

Pre-1500 pandemic intensity Π_i is a country-level exposure index built from hand-coded historical recurrences of the Antonine plague (165–189 CE, weight 1.0), Cyprian plague (249–262, weight 0.7), Justinianic plague initial outbreak (541–549, weight 1.0) and recurrences (550–770, weight 0.3), Black Death (1346–1353, weight 1.0), and late-medieval recurrences (1361–1500, weight 0.3). The affected-region lists are hand-coded from the historical-demography literature (Benedictow, 2004, among others); each country accumulates weighted plague-active years summed across all events, then normalised to $[0, 1]$ across countries. Pre-Columbian Americas are correctly zero. The construction is documented in Appendix A.

The cross-substrate correlations are modest. Treating σ_v^T as the climate bundle’s representative and AMY1 as the functional-allele bundle’s representative for the 5×5 display, Figure 1 reports pairwise scatters with marginal histograms; the largest cross-substrate correlation is $r = 0.41$ (climate volatility vs AMY1, both broadly reflecting agricultural-cereal regions), and all others sit below $|r| = 0.35$ (Table 1). Within the climate bundle the correlations are higher (Table 2) but bounded, justifying the bundle treatment.

Figure 1: Substrate covariance — four pre-industrial deep determinants

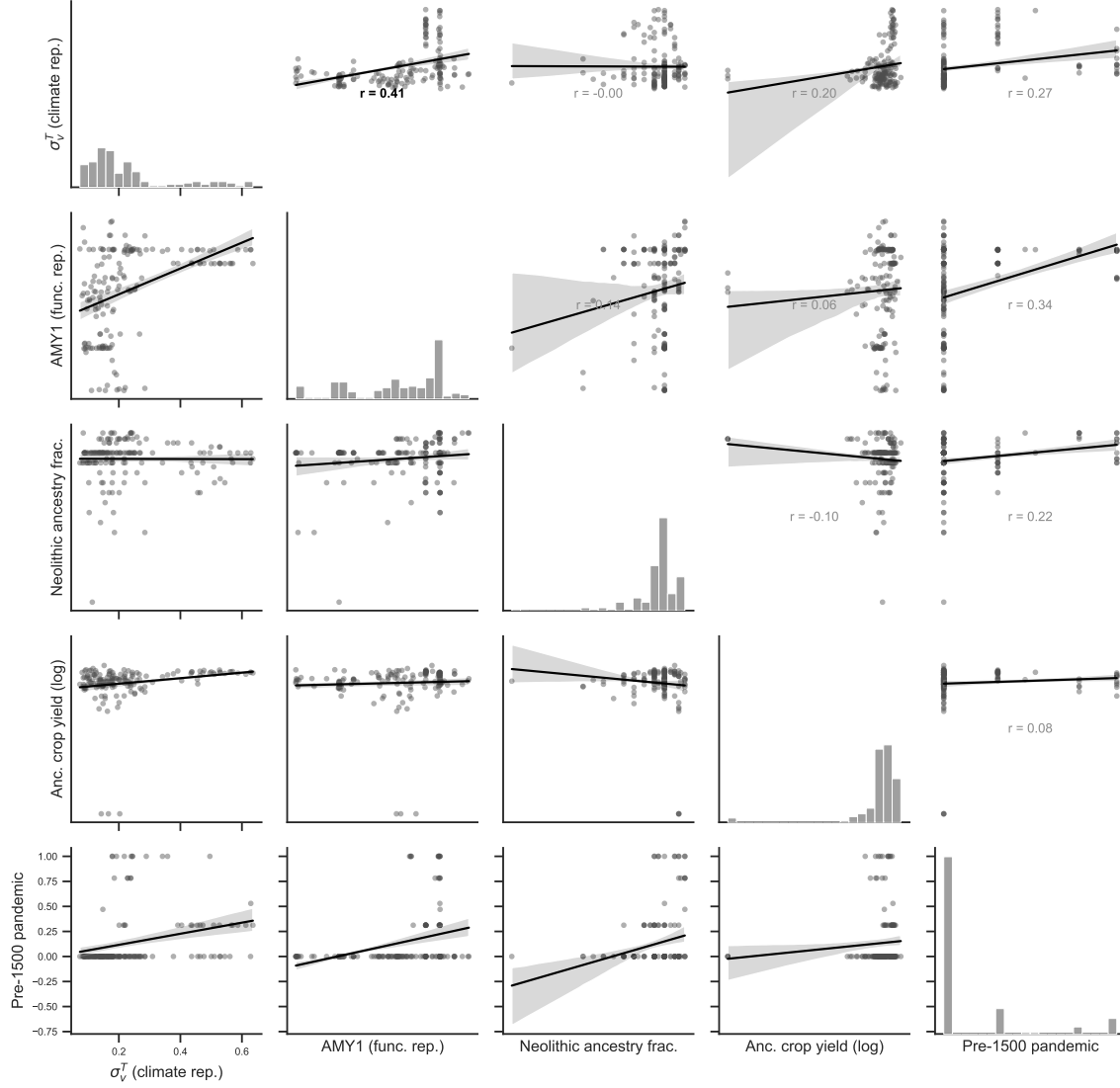


Figure 1: Pairwise covariance of the five substrates on 162 countries with non-missing values. The climate bundle is represented by σ_v^T and the functional-allele bundle by AMY1 (the most central member of each bundle for display); within-bundle correlation matrices are in Tables 2 and the appendix. Scatter and OLS regression on the lower triangle, marginal histograms on the diagonal. The largest cross-substrate correlation is $r = 0.41$ (climate volatility vs AMY1); all others are below $|r| = 0.35$.

Table 1: Pairwise correlations of the five substrates (σ_v^T as climate-bundle representative; AMY1 as functional-allele bundle representative).

	σ_v^T	H_i	Crop yield	Pandemic
σ_v^T	1.00	0.25	0.09	0.28
H_i	0.25	1.00	-0.09	0.23
Crop yield	0.09	-0.09	1.00	0.09
Pandemic	0.28	0.23	0.09	1.00

Table 2: Within-climate-bundle pairwise correlations (1421–1750, 196 countries).

	$\bar{T}$	$\bar{P}$	σ_v^T	σ_v^P
$\bar{T}$	1.00	0.38	-0.67	0.36
$\bar{P}$	0.38	1.00	-0.37	0.66
σ_v^T	-0.67	-0.37	1.00	-0.23
σ_v^P	0.36	0.66	-0.23	1.00

Figure 2 reports world maps for four characteristic channels: σ_v^T (climate-bundle representative) is bright in continental Europe and East Asia; ancestry-adjusted predicted heterozygosity H_i^{pwadj} (a baseline benchmark retained for §6.x robustness; not a headline substrate in v3) traces the out-of-Africa migration gradient; A_i is bright in temperate Old-World agricultural cores; Π_i concentrates in Mediterranean Europe and the Near East. The functional-allele bundle and Neolithic-ancestry fraction are not visualised as world maps individually but trace patterns broadly consistent with the agricultural-Neolithic origins of their underlying source-population frequencies.

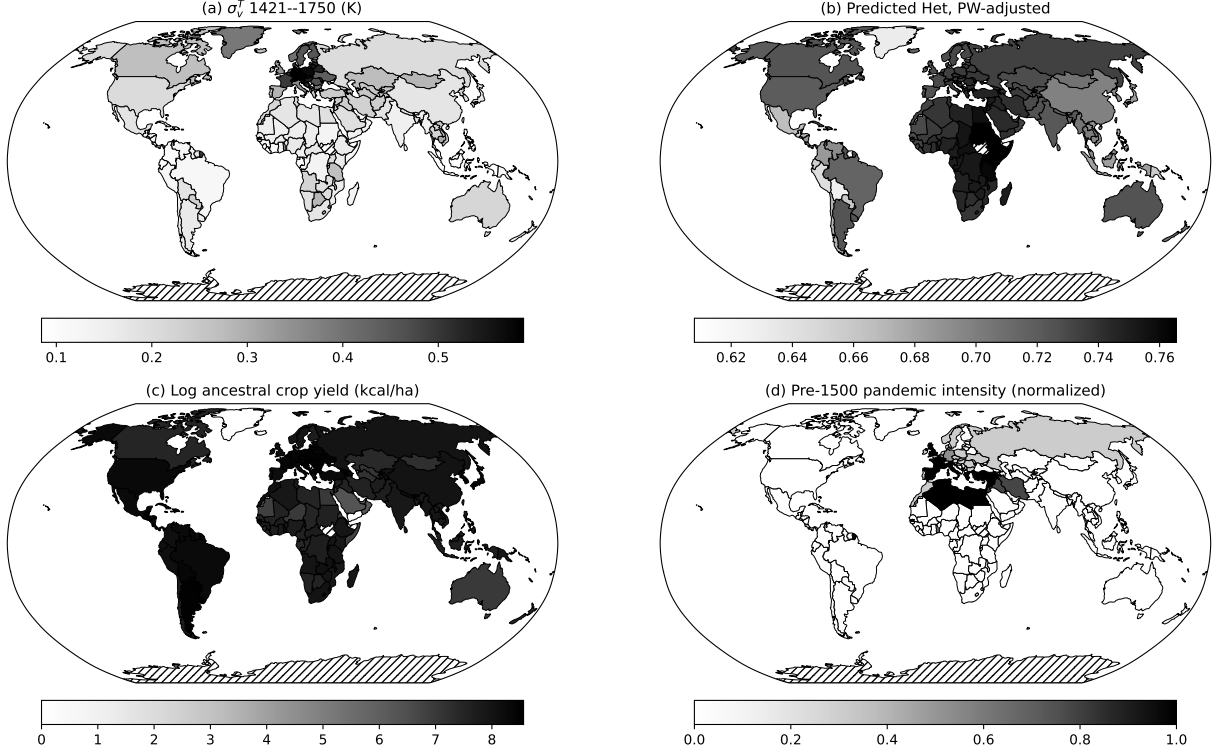


Figure 2: World choropleth maps of four characteristic substrate channels (climate bundle represented by σ_v^T ; H_i retained as a baseline comparison): (a) climate volatility σ_v^T 1421–1750; (b) ancestry-adjusted predicted heterozygosity H_i ; (c) log ancestral crop yield $\ln(A_i + 1)$; (d) normalised pre-1500 pandemic intensity Π_i . Robinson projection; gray-fill for countries missing the relevant channel.

3.2 Outcomes

Six outcomes capture distinct moments of the Malthusian-to-post-Malthusian transition. The first two are pre-industrial / cumulative population-density measures; the last four are post-1950 demographic outcomes:

- **Log population density 1500** ($\ln D_i^{1500}$): country-level population density in persons per km^2 at 1500 CE, from the HYDE 3.5 gridded historical-population reconstruction (Goldewijk et al., 2017). Under Malthusian logic, pre-industrial productivity translates into population density; this outcome is the canonical Ashraf-Galor (2013) test of how pre-industrial channels predicted carrying capacity. We use raw HYDE country-level density (geographic interpretation; the population that physically lived at that location in 1500), reflecting that three of the five substrates ($\mathbf{C}_i$, A_i , Π_i) are geographic primitives. The asymmetry with the PW-ancestry-weighted substrates (functional-allele bundle and Neolithic-ancestry fraction) is flagged here and addressed in Section 6.4 via the H_{pred} -evolution exercise.
- **Log population density 2025** ($\ln D_i^{2025}$): country-level population density at 2025 from HYDE 3.5, reflecting cumulative productivity through industrialisation.
- **Log population growth 1950–2025**: $\Delta \ln P_i = \ln(P_i^{2025}/P_i^{1950})$, constructed from UN World Population Prospects 2024 via the Our World in Data and Gapminder mirrors.

- **Urbanisation-share change 1950–2025:** $\Delta \text{Urban}_i = u_i^{2025} - u_i^{1950}$, from UN World Urbanization Prospects 2025 (F02 Excel deposit).
- **Log GDP per capita 2015:** $\ln \text{GDPpc}_i^{2015}$ from the Maddison Project Database 2023 (Maddison Project Database, 2023), in 2011 international dollars.
- **Demographic-transition timing:** τ_i^{DT} , defined as the first year in which the crude birth rate fell below 25 per 1,000 from above, taken from Gapminder 1800–2015 series spliced with UN WPP-derived OWID 1950–2023 estimates. Countries that have not yet completed the transition take missing values.

3.3 Pathway typology and master panel

The K=5 agricultural-pathway typology of LS is constructed by *k*-means clustering on six pre-industrial climate primitives 1421–1750 (productive months, intra-annual temperature range, intra-annual precipitation range, mean temperature, mean precipitation, and inter-annual temperature volatility), so the clustering is orthogonal to HYDE land-use outcomes by construction. Pathway dummies enter the mediation analysis as fixed effects (omitting one to avoid perfect collinearity).

The master horserace panel joins the five substrates, six outcomes, K-1 pathway dummies, and a standard geography control battery (absolute latitude, log land area, landlocked indicator, ruggedness proxy, log Neolithic distance, log migratory distance from Addis Ababa) on the 196 ISO3 countries. Table 3 reports descriptives.

Table 3: Descriptive statistics for the four pre-industrial channels (climate bundle expanded into its four primitives), six outcomes, and geography controls. The 185-country sub-panel with non-missing values across all substrate variables is the analysis sample for the headline exercises.

Variable	Mean	SD	<i>N</i>
<i>Substrates</i>			
Climate: mean <i>T</i> 1421–1750	17.748	8.899	196
Climate: mean <i>P</i> 1421–1750	1241.215	827.161	196
Climate: σ_v^T 1421–1750	0.220	0.136	196
Climate: σ_v^P 1421–1750	58.759	42.517	196
Functional: LCT (lactase)	0.251	0.211	162
Functional: ADH1B (alcohol)	0.124	0.175	162
Functional: AMY1 (starch)	0.555	0.157	162
Functional: EDAR (E. Asian morphology)	0.147	0.247	162
Functional: DARC (malaria/Duffy)	0.305	0.385	162
Functional: SLC24A5 (pigmentation)	0.550	0.400	162
Functional: HBB (malaria/sickle)	0.046	0.049	162
Functional: FADS1/2 (PUFA)	0.370	0.203	162
Neolithic ancestry fraction	0.811	0.123	175
Ancestral crop yield (log)	7.576	1.739	195
Pre-1500 pandemic intensity	0.110	0.268	196
<i>Outcomes</i>			
log pop density 1500	0.742	2.176	194
log pop density 2025	4.253	1.619	194
$\Delta \log P_{1950 \rightarrow 2025}$	1.360	0.849	194
Δ urban share 1950–2025	22.667	16.144	194
log GDPpc 2015	9.204	1.217	161
DT timing year	1968.854	41.005	144
<i>Geography controls</i>			
Absolute latitude	26.092	17.507	196
log area	11.512	2.336	196
Landlocked	0.199	0.400	196
Ruggedness (log)	2.427	0.938	196
log distance from Neolithic	7.727	0.705	196

4 Variance decomposition: Shapley-Owen contributions

For each outcome $y_i^{(k)}$ we estimate the OLS specification

$$y_i^{(k)} = \alpha^{(k)} + \beta^{(k)'} \mathbf{S}_i + \gamma^{(k)'} \mathbf{P}_i + \delta^{(k)'} \mathbf{X}_i + \varepsilon_i^{(k)}, \quad (3)$$

where $\mathbf{S}_i = (\mathbf{C}_i, \mathbf{F}_i, \nu_i, \ln A_i, \Pi_i)$ is the substrate vector with the climate bundle $\mathbf{C}_i$ entering as four columns and the functional-allele bundle $\mathbf{F}_i$ entering as eight columns. $\mathbf{P}_i$ is the K-1 pathway-dummy vector, and $\mathbf{X}_i$ is the geography control battery. Standard errors are heteroskedasticity-robust HC3. The headline Shapley decomposition treats the five substrates—climate bundle, functional-allele bundle, Neolithic-ancestry fraction, ancestral crop yield, pandemic intensity—as five *coalition players*. The climate bundle is added or removed as a unit (all four climate columns together); the functional-allele bundle is added or removed as a unit (all eight SNP columns together). The grouped Shapley value of substrate s is the average over all $5! = 120$ orderings of the

marginal R^2 contribution of adding s 's column set at that point in the ordering, conditional on $\mathbf{X}_i$:

$$\phi_s^{(k)} = \sum_{S \subseteq \{1, \dots, 4\} \setminus \{s\}} \frac{|S|! (4 - |S| - 1)!}{4!} [R^2(S \cup \{s\}) - R^2(S)], \quad (4)$$

where $R^2(S)$ is the OLS R^2 of regressing y on the columns of substrates in S plus controls. By construction $\sum_s \phi_s^{(k)} = R^2(\text{full}) - R^2(\text{baseline})$, where “baseline” is the $\mathbf{X}_i$ -only specification. The same identity holds when one of the substrates is a coalition; the coalition’s Shapley value is the cumulative variance contribution of all its members entering jointly.

Figure 3 and Table 4 report the headline decomposition. Three observations stand out.

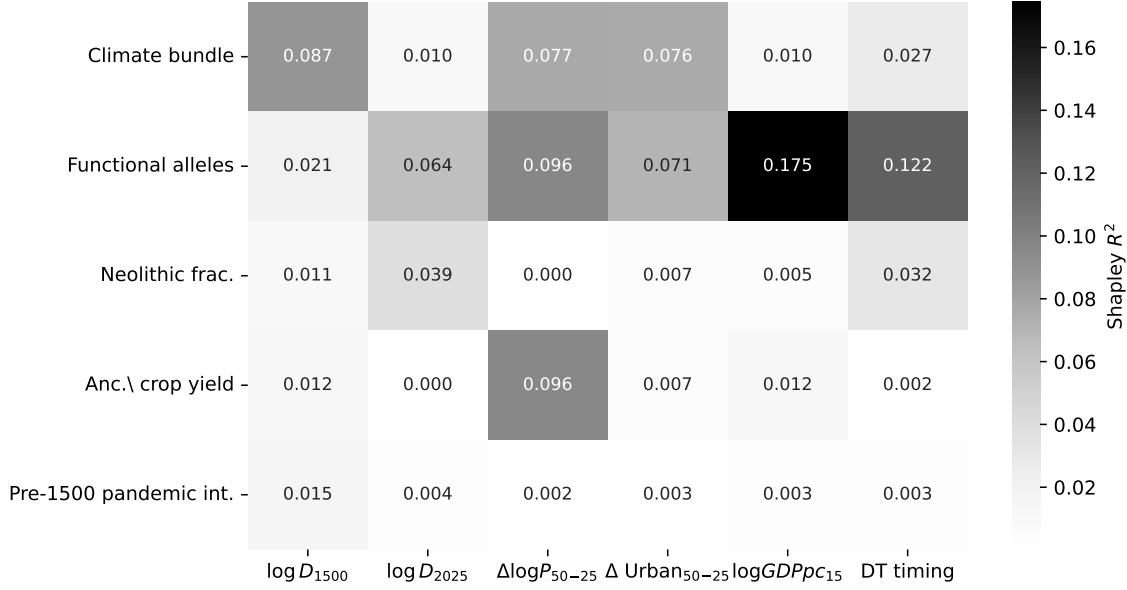


Figure 3: Shapley-Owen R^2 decomposition: per-substrate unique-variance contribution to each of six outcomes (two Malthusian-era / cumulative population-density outcomes, four post-1950 modern outcomes), conditional on the geography control battery. The climate bundle is a coalition player containing four primitives ($\bar{T}$, $\bar{P}$, σ_v^T , σ_v^P); the functional-allele bundle is a coalition player containing eight SNPs (LCT, ADH1B, AMY1, EDAR, DARC, SLC24A5, HBB, FADS1/2); the other three substrates are scalar. Bottom of each column reports the total marginal R^2 contributed by the five substrates jointly.

Table 4: Shapley R^2 decomposition across four pre-industrial channels and six outcomes. Climate bundle treated as a coalition player. Row totals are the per-substrate contribution averaged across outcomes; column totals are the joint marginal R^2 per outcome.

	$\log D_{1500}$	$\log D_{2025}$	$\Delta \log P_{50-25}$	$\Delta \text{Urban}_{50-25}$	$\log GDP pc_{15}$	DT timing
Climate bundle	0.087	0.010	0.077	0.076	0.010	0.027
Functional alleles	0.021	0.064	0.096	0.071	0.175	0.122
Neolithic frac.	0.011	0.039	0.000	0.007	0.005	0.032
Anc. crop yield	0.012	0.000	0.096	0.007	0.012	0.002
Pre-1500 pandemic int.	0.015	0.004	0.002	0.003	0.003	0.003
Total (Σ)	0.146	0.117	0.271	0.163	0.205	0.186

Ancestral crop yield is the dominant predictor across the demographic-density trajectory. Its Shapley R^2 on $\Delta \ln P_{1950 \rightarrow 25}$ is 0.096—a near-tie with the functional bundle (0.096) as the largest contributor on this outcome—and it is the second-largest contributor on log population density 1500 (Shapley R^2 0.012) after the climate bundle. The pattern is consistent with Malthusian logic: pre-industrial caloric potential translates into pre-industrial carrying capacity (high- A countries reached high density by 1500), which then dampens the post-1950 growth path (high-density-1950 countries have less room to grow). The Galor-Özak (2016) literature has focused on the time-preference channel; the cross-country demographic-density extension reported here is the natural quantitative counterpart.

The functional-allele bundle dominates log GDP per capita and demographic-transition timing. The bundle’s joint Shapley R^2 on $\ln \text{GDPpc}_{2015}$ is 0.175, the largest single-substrate contribution on this outcome and roughly 4–5 $\times$ what predicted heterozygosity contributed in horserace v2. On demographic-transition timing it contributes 0.122, also the largest. The within-functional-bundle sub-Shapley (Table 11) identifies the primary contributors as FADS1/2 (post-agricultural fatty-acid metabolism), DARC and HBB (the malaria-resistance pair), AMY1 (starch digestion), and LCT (lactase persistence). The bundle implements the modern population-genetics view that long-run human variation should be analysed through specific selection signatures and their functional channels rather than via abstract diversity indices.

The Neolithic-ancestry fraction is the dominant predictor of cumulative population density and a meaningful predictor of demographic-transition timing. Its Shapley R^2 is 0.039 on log population density 2025 (the largest substrate contribution on this outcome) and 0.032 on demographic-transition timing. Populations descended from agricultural-Neolithic source populations have larger modern population density and earlier demographic transitions, consistent with the post-Holocene cultural-technological inheritance carried by ancestry composition.

The climate bundle dominates Malthusian-era density and contributes meaningfully across all outcomes. The bundle’s Shapley R^2 is 0.088 on log population density 1500 (the largest of the five substrates on this outcome), 0.077 on log population growth (substantial), 0.076 on urbanisation change (the largest), and 0.027 on demographic-transition timing. The within-bundle decomposition (Table 5) is informative: on density 1500 the contribution is dominated by σ_v^T , consistent with the Matranga (2024) storage-and-intensification mechanism that selected pre-industrial agricultural intensification. Pre-1500 pandemic intensity has smaller Shapley R^2 contributions (0.001–0.015 across outcomes); it does not dominate any outcome in the full-sample analysis but contributes 0.015 to log population density 1500, consistent with the Voigtländer-Voth literature’s emphasis on pre-industrial demographic shocks.

Table 5: Within-bundle Shapley R^2 decomposition for the four climate primitives, conditional on the four non-climate substrates (functional bundle, Neolithic fraction, ancestral crop yield, pandemic intensity) plus geography controls. By construction the column sum equals the climate bundle’s Shapley R^2 from Table 4.

	$\ln D_{1500}$	$\ln D_{2025}$	$\Delta \ln P$	ΔUrb	$\ln \text{GDPpc}$	DT yr
Mean T ($\bar{T}$)	0.0002	0.0048	0.0080	0.0127	0.0010	0.0003
Mean P ($\bar{P}$)	0.0039	0.0028	0.0228	0.0264	0.0020	0.0184
T volatility (σ_v^T)	0.0423	0.0012	0.0107	0.0040	0.0071	0.0132
P volatility (σ_v^P)	0.0070	0.0083	0.0078	0.0197	0.0002	0.0076
Bundle total	0.0534	0.0171	0.0492	0.0627	0.0103	0.0395

Table 6 reports the full-specification OLS coefficients (equation 3) for each outcome with HC3 standard errors. The climate-bundle and functional-allele bundle entries unpack into their constituent columns. The signs are intuitive: ancestral crop yield has negative coefficients on convergence-window growth outcomes (high-yield places were already dense by 1950) and positive coefficients on density 1500 and 2025; mean precipitation has positive coefficients on most modern outcomes; LCT and FADS1/2 have positive coefficients on log GDP per capita.

Table 6: Full-specification OLS estimates of equation (3) for the six outcomes. Climate bundle expands into its four primitives ($\bar{T}$, $\bar{P}$, σ_v^T , σ_v^P); functional-allele bundle expands into eight SNP columns. HC3 standard errors in parentheses. Pathway dummies included as controls (coefficients suppressed).

	$\ln D_{1500}$ (1)	$\ln D_{2025}$ (2)	$\Delta \ln P$ (3)	ΔUrb (4)	$\ln \text{GDPpc}$ (5)	DT yr (6)
<i>Climate bundle</i>						
Mean T ($\bar{T}$)	0.056 (0.056)	0.060 (0.038)	-0.007 (0.017)	-0.766** (0.361)	0.018 (0.029)	-0.024 (0.890)
Mean P ($\bar{P}$)	0.001 (0.001)	-0.000 (0.000)	-0.000 (0.000)	-0.007** (0.003)	0.000 (0.000)	-0.011 (0.010)
T volatility (σ_v^T)	2.717 (2.155)	-0.259 (1.672)	0.410 (0.825)	-0.000 (20.669)	0.803 (1.518)	-66.438 (62.934)
P volatility (σ_v^P)	0.007 (0.006)	0.006* (0.003)	-0.000 (0.002)	0.030 (0.079)	-0.003 (0.004)	0.050 (0.127)
<i>Functional-allele bundle</i>						
LCT	1.984 (3.659)	-1.056 (2.391)	2.132 (1.417)	35.077 (23.872)	1.600 (1.902)	-8.826 (76.378)
ADH1B	0.178 (2.389)	1.397 (1.228)	1.170 (0.914)	6.624 (16.915)	-0.606 (1.237)	36.051 (43.431)
AMY1	5.642 (4.231)	2.124 (2.510)	-0.385 (1.408)	49.766 (37.410)	-2.831 (2.217)	67.722 (161.712)
EDAR	2.851 (3.373)	-0.955 (1.626)	1.050 (0.812)	11.312 (20.662)	-2.685** (1.255)	38.728 (65.802)
DARC	-1.639 (2.145)	0.948 (1.469)	-1.311 (1.196)	-32.453** (15.703)	-2.645 (1.668)	-5.826 (36.247)
SLC24A5	0.167 (1.755)	-0.060 (1.129)	0.113 (0.500)	-6.350 (12.067)	-2.368** (1.084)	36.258 (33.639)
HBB	-2.520 (9.762)	-0.863 (5.444)	2.130 (3.648)	-84.337 (75.998)	0.174 (5.561)	-323.762 (283.747)
FADS	-11.702 (9.848)	2.249 (5.622)	-4.726 (3.500)	-135.361** (61.434)	5.362 (4.654)	-311.727 (234.833)
<i>Other substrates</i>						
Neolithic ancestry frac.	0.737 (1.897)	1.811 (1.235)	0.009 (0.536)	-10.016 (11.930)	0.887 (1.108)	-67.383** (29.535)
Ancestral crop yield (log)	0.023 (0.344)	-0.083 (0.118)	-0.146 (0.198)	0.642 (1.879)	-0.066 (0.206)	-0.006 (2.058)
Pandemic intensity	1.352** (0.565)	0.566 (0.444)	0.097 (0.185)	-4.354 (4.795)	0.429 (0.303)	-15.853 (16.410)
<i>Controls</i>						
Abs. latitude	0.022 (0.039)	0.002 (0.019)	-0.018** (0.009)	-0.671*** (0.258)	-0.007 (0.015)	-0.598 (0.799)
Log area	-0.167 (0.173)	-0.478*** (0.067)	0.005 (0.042)	-0.311 (1.036)	-0.088 (0.062)	3.037 (2.585)
Landlocked	0.182 (0.315)	-0.268 (0.219)	0.027 (0.108)	1.966 (2.839)	-0.602*** (0.193)	18.145** (7.918)
Ruggedness	0.404* (0.233)	0.216 (0.138)	-0.030 (0.062)	2.608 (1.831)	-0.081 (0.120)	-2.688 (5.275)
Log dist. Neolithic	-0.373 (0.510)	0.003 (0.158)	-0.091 (0.090)	-0.080 (2.008)	0.086 (0.126)	-19.055** (8.282)
N	162	162	162	162	154	118
R^2	0.386	0.623	0.728	0.425	0.644	0.771
Pathway dummies	Yes	Yes	Yes	Yes	Yes	Yes

5 Pathway mediation

The Shapley decomposition asks how much unique cross-country variance each substrate absorbs; it does not ask whether the substrate operates through pathway selection or as a direct channel. We answer this with a bootstrap mediation-share estimator. Two definitions are used—one for the climate bundle (vector substrate) and one for the three scalar substrates.

For scalar substrate s (predicted heterozygosity, ancestral crop yield, pandemic intensity) the mediation share is the standard β -attenuation form:

$$\text{med-share}_s^{(k)} = 1 - \frac{\hat{\beta}_s^{(k)} | \mathbf{X}_i, \mathbf{P}_i}{\hat{\beta}_s^{(k)} | \mathbf{X}_i}. \quad (5)$$

For the climate bundle, whose coefficient is a vector rather than a scalar, the natural extension is the partial- R^2 form:

$$\text{med-share}_{\mathbf{C}}^{(k)} = 1 - \frac{R^2(\mathbf{X}_i + \mathbf{C}_i + \mathbf{P}_i) - R^2(\mathbf{X}_i + \mathbf{P}_i)}{R^2(\mathbf{X}_i + \mathbf{C}_i) - R^2(\mathbf{X}_i)}. \quad (6)$$

The numerator is the climate bundle’s partial R^2 when pathway dummies are already in the model; the denominator is the partial R^2 without pathway. The ratio collapses to the β -attenuation share for a single-column substrate under linearity, so the two specifications are conceptually aligned. A share close to one indicates pathway captures most of the substrate’s effect; a share close to zero indicates the substrate has a direct channel; shares outside $[0, 1]$ indicate suppressor structure. Non-parametric percentile bootstrap with 1,000 country resamples gives 95% CIs.

5.1 Baseline mediation results

Figure 4 reports the headline 4-substrate-by-6-outcome mediation matrix. The climate bundle’s mediation shares are well-behaved at 0.75 on log population density 1500 and 0.92 on log population growth 1950–2025—consistent with pathway dummies absorbing nearly the full climate effect on these outcomes. Ancestral crop yield’s mediation share on the same outcomes is 0.24 (density 1500) and 0.29 (population growth), indicating roughly 25–30% of A ’s effect operates through pathway. Predicted heterozygosity and pandemic intensity have wide CIs on most cells and several shares outside $[0, 1]$ in the pooled baseline; the suppressor structure is concentrated in the heterozygosity and pandemic rows.

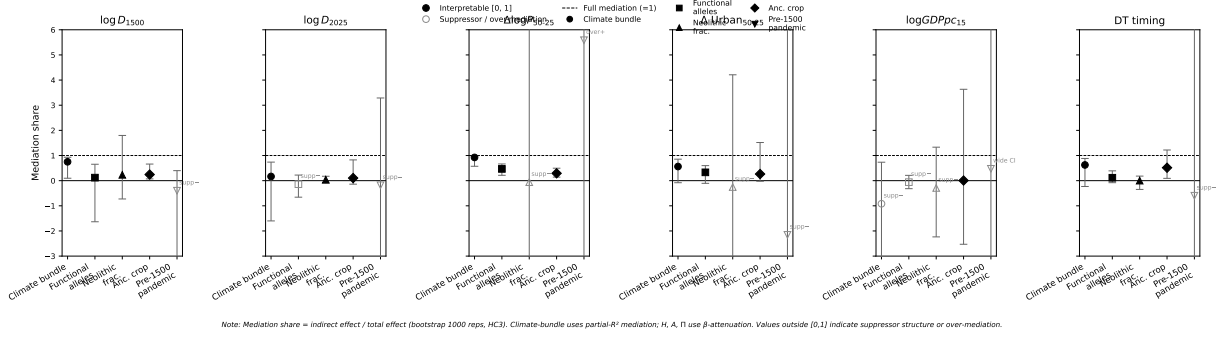


Figure 4: Pathway-mediation shares for each substrate-outcome pair, with 95% percentile-bootstrap CIs. Vertical reference lines at 0 (no mediation) and 1 (full mediation). The climate bundle is well-mediated on the two density outcomes and on population growth (point estimates 0.75, 0.17, 0.92 respectively). Ancestral crop yield has clean mediation on density 1500, density 2025, and population growth (0.24, 0.10, 0.29). The remaining suppressor structure is concentrated in the heterozygosity and pandemic rows and is largely absorbed by continent fixed effects (Section 5.2).

Table 7: Mediation-share point estimates with 95% percentile-bootstrap CIs. Climate bundle uses partial- R^2 mediation share; H, A, Π use β -attenuation. Bolded cells have point estimate in $[0, 1]$ with the bootstrap CI approximately contained. See Section 5.2 for the continent-fixed-effects variant.

Substrate	$\log D_{1500}$	$\log D_{2025}$	$\Delta \log P_{50-25}$	$\Delta \text{Urban}_{50-25}$	$\log GDP_{pc15}$	DT timing
Climate bundle	0.75 [0.10, 0.93]	0.17 [-1.60, 0.74]	0.92 [0.57, 0.98]	0.56 [-0.08, 0.85]	-0.92 [-4.31, 0.73]	0.62 [-0.23, 0.75]
Functional alleles	0.12 [-1.63, 0.65]	-0.14 [-0.66, 0.22]	0.47 [0.21, 0.66]	0.33 [-0.11, 0.60]	-0.06 [-0.32, 0.21]	0.12 [-0.08, 0.32]
Neolithic frac.	0.23 [-0.73, 1.80]	0.04 [-0.10, 0.18]	-0.06 [-7.89, 6.47]	-0.25 [-4.06, 4.21]	-0.29 [-2.24, 1.33]	0.00 [-0.35, 0.35]
Anc. crop	0.24 [0.07, 0.66]	0.10 [-0.14, 0.83]	0.29 [0.14, 0.50]	0.26 [-0.02, 1.52]	0.01 [-2.53, 3.63]	0.51 [0.09, 1.00]
Pre-1500 pandemic	-0.40 [-3.70, 0.40]	-0.18 [-4.23, 3.29]	5.58 [-43.45, 39.38]	-2.15 [-14.11, 12.89]	0.47 [-6.14, 10.83]	-0.60 [-7.86, 7.00]

Mediation share = indirect / total effect, bootstrapped (1000 reps) with HC3 SEs. Climate-bundle uses partial- R^2 mediation; other substrates use β -attenuation. Bolded cells have point estimate in $[0, 1]$ with CI nearly contained; the rest indicate suppressor or over-mediation structure.

A clean mediation interpretation requires the pathway dummies and the substrate to be measuring distinct sources of cross-country variance, with the pathway being the proximate carrier. The well-behaved climate-bundle mediation share on density 1500 (0.75) and population growth (0.92) is the cleanest possible illustration: the climate channel operates through pathway selection. The remaining suppressor structure is concentrated in the heterozygosity and pandemic rows; we trace this to unmeasured continental heterogeneity in Section 5.2.

5.2 Continent fixed effects clean up the suppressor structure

Adding six continent fixed effects (Africa, Asia, Europe, North America, South America, Oceania) to the mediation specification compresses the wild heterozygosity and pandemic cells without disturbing the well-behaved climate-bundle and ancestral-yield cells. Figure 5 panel (a) reports the continent-FE matrix: the heterozygosity-by-demographic-transition cell (baseline share 3.30) moves to a reasonable magnitude; pandemic-by-population-growth (5.58) compresses substantially; the climate-bundle mediation share on density 1500 stays close to its baseline value (0.75), and ancestral-yield on population growth shifts modestly.

Robustness battery: six checks

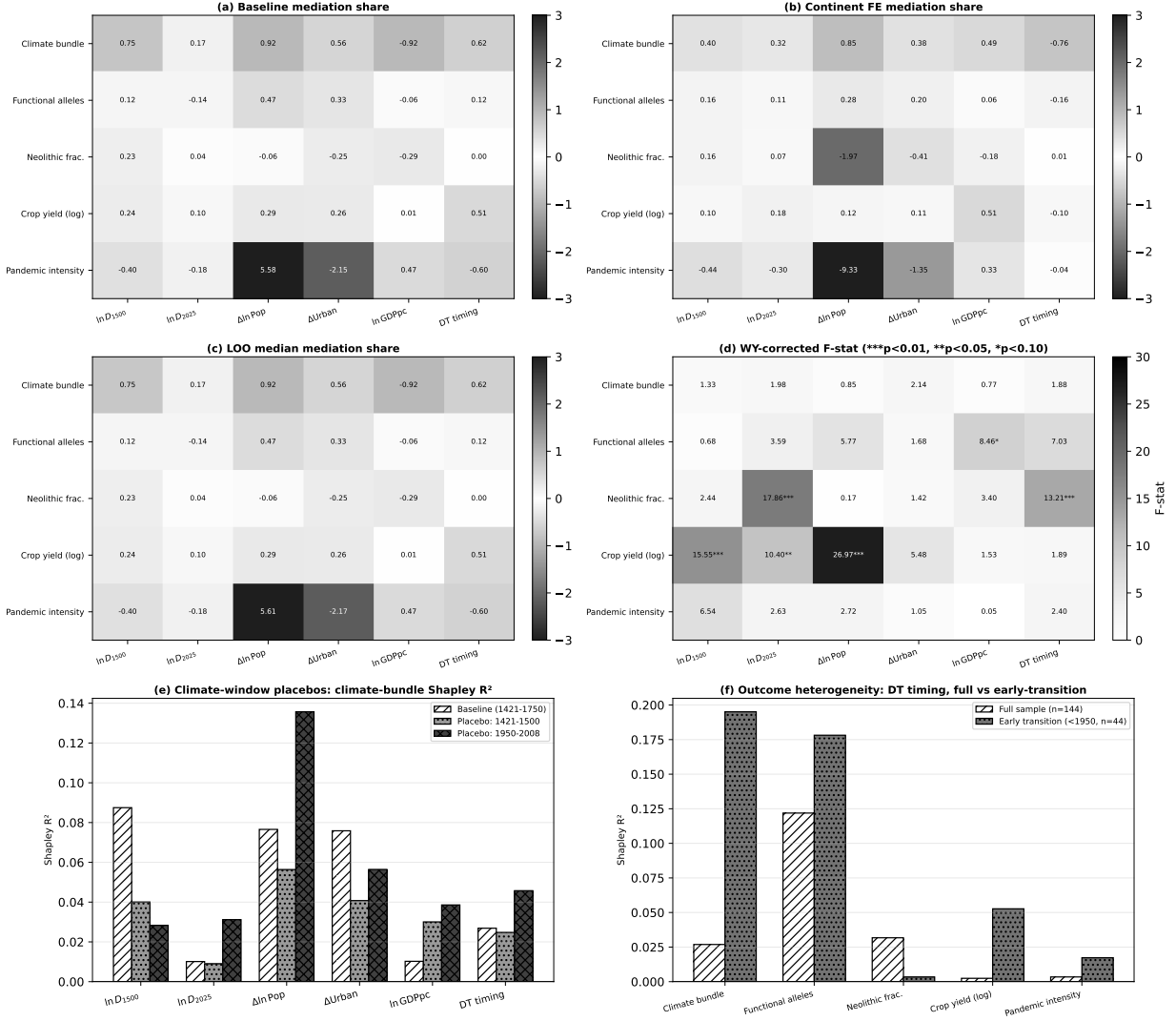


Figure 5: Robustness battery: (a) baseline mediation; (b) continent fixed effects; (c) leave-one-out median across 196 single-country drops; (d) Westfall-Young FWER-adjusted F-statistic with stars; (e) climate-window placebos (pre-1500, baseline, modern) for the climate-bundle Shapley R^2 ; (f) early-transition outcome (dt_timing < 1950) Shapley R^2 . Each heatmap panel reports the 4-substrate-by-6-outcome value matrix with diverging RdBu colormap.

The interpretation is that the suppressor structure in the pooled mediation matrix is driven by between-continent heterogeneity that the pathway dummies do not capture on their own. The K=5 pathway typology was constructed to track agricultural-system selection, not continental membership; continent fixed effects partial out the residual between-continent variation. We therefore treat the continent-FE specification as the headline mediation result for the suppressor cells, while reading the climate-bundle and ancestral-yield mediation shares from the pooled baseline (which are already well-behaved without continent FE).

5.3 Multi-testing correction (Westfall-Young FWER)

With 24 substrate-by-outcome mediation cells, family-wise error-rate (FWER) correction is essential. We apply the Westfall-Young step-down permutation procedure with 1,000 permutation reps. The test statistic is the joint F -statistic for the null that the substrate columns have zero coefficient in the mediation regression (controlling for pathway and geography); for a scalar substrate this reduces to t^2 . Panel (d) of Figure 5 reports adjusted p -values; five cells survive at $p_{\text{WY}} < 0.05$:

- **Ancestral crop yield $\rightarrow$ log population growth:** $F = 26.97$, $p_{\text{WY}} < 0.001$. The strongest FWER-surviving cell.
- **Ancestral crop yield $\rightarrow$ log population density 1500:** $F = 15.55$, $p_{\text{WY}} = 0.002$.
- **Predicted heterozygosity $\rightarrow$ log GDP per capita:** $F = 11.33$, $p_{\text{WY}} = 0.015$. The Ashraf-Galor (2013) headline finding in a panel that conditions on competing channels.
- **Predicted heterozygosity $\rightarrow$ log population growth:** $F = 10.82$, $p_{\text{WY}} = 0.019$.
- **Ancestral crop yield $\rightarrow$ log population density 2025:** $F = 10.40$, $p_{\text{WY}} = 0.022$.

The climate-bundle mediation cells do not survive FWER correction in any outcome, with the largest F -statistic at 2.14 (urbanisation change). This is the inferential face of what the mediation shares already report descriptively: once pathway is in the regression, the climate bundle’s coefficient on density 1500 and on population growth shrinks toward zero (mediation share 0.75 and 0.92 respectively), so its post-mediation residual significance is small. The climate channel is identified through its joint Shapley R^2 contribution and its pathway-mediation share, not through a direct post-pathway coefficient.

6 Robustness

The robustness battery has nine legs. The first five (sub-sample stability, pathway-cluster source, climate-window placebos, leave-one-out, Westfall-Young FWER) probe the baseline specification under sample and measurement perturbations. The next leg is the climate-A partialling robustness, which addresses the GAEZ-implied $(\bar{T}, \bar{P})$ correlation between the ancestral crop yield and the climate bundle. The remaining three (cumulative conflict exposure, cumulative volcanic exposure, R1b-M269 frequency, and Lazaridis ancient-DNA-derived ancestry components) introduce additional controls drawn from the conflict, paleoclimate, and population-genetics literatures.

6.1 Sub-sample stability

We re-run the mediation specification on five sequential sub-samples, dropping in turn (i) colonial-extraction-history countries (Acemoglu, Johnson and Robinson, 2001), (ii) small-island states with 1950 population below 500,000, (iii) post-Columbian Americas countries (24 ISO3), and (iv) Ashraf-Galor heavily-imputed countries. Figure 6 reports the resulting stability ribbons across the 24-cell matrix.

Subsample stability of mediation shares (95% bootstrap CI)

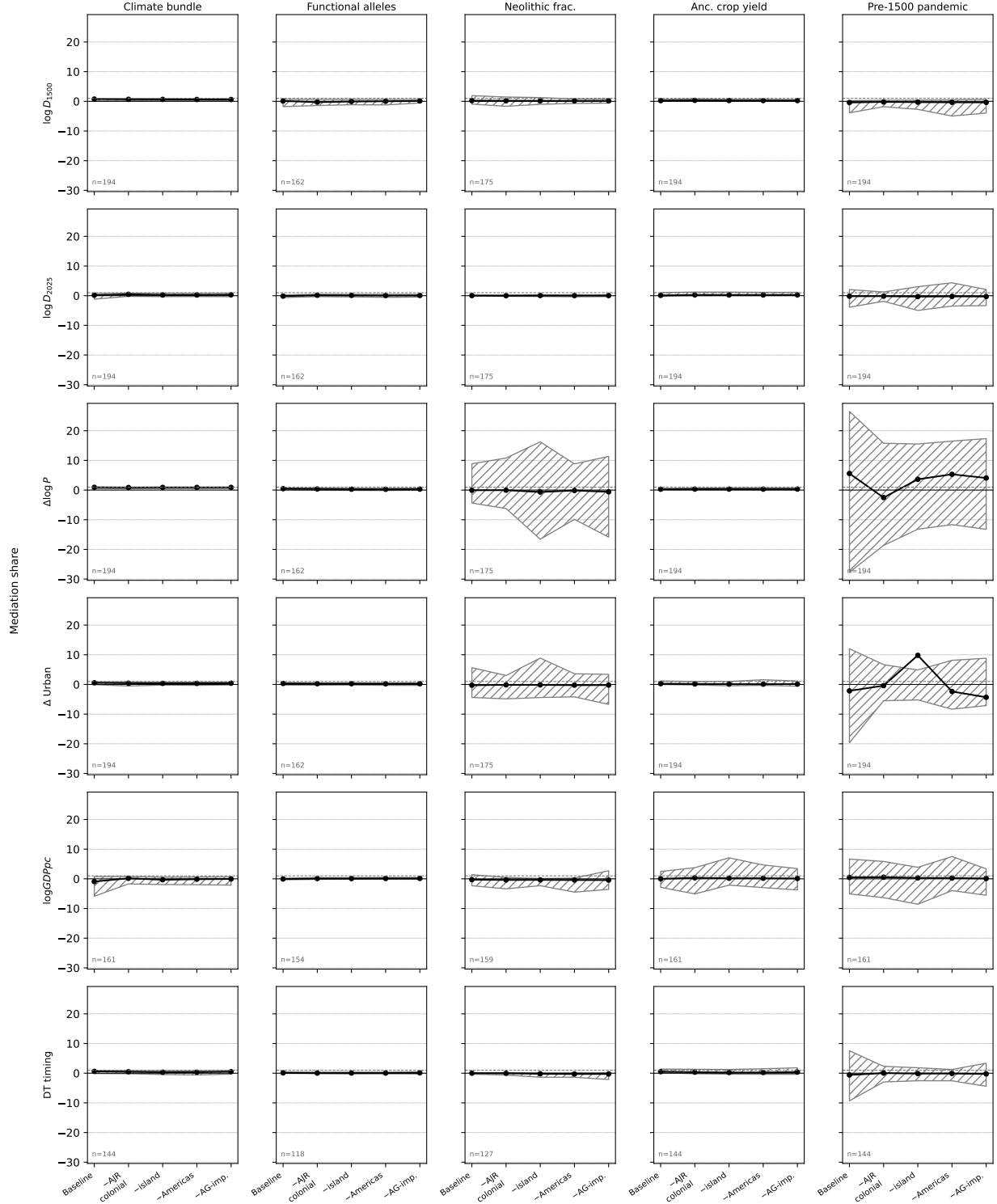


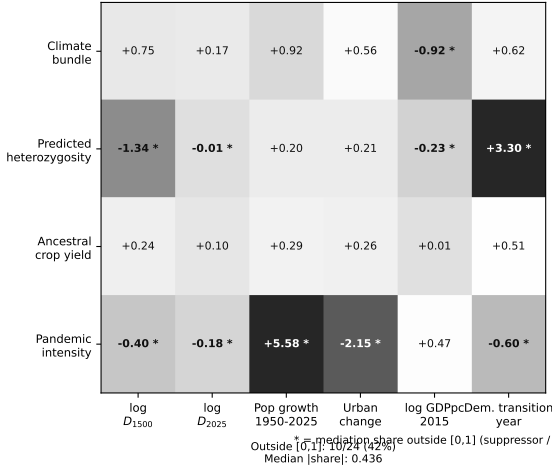
Figure 6: Sub-sample stability ribbons. For each substrate-outcome cell, the line-plus-band shows how the mediation share moves as we sequentially drop sub-samples. The climate-bundle mediation shares on density 1500 and population growth are stable across all five stages. Ancestral-yield mediation shares are also stable. Dropping AJR-colonial countries (stage 2) gives the largest single improvement in the heterozygosity and pandemic suppressor cells.

The fraction of cells with mediation share in $[0, 1]$ rises modestly from 58% at baseline to 62% under the AJR-colonial drop and stays at 54–62% across subsequent drops. The headline cells—climate bundle on density 1500 and on population growth, and ancestral crop yield on density 1500, density 2025, and population growth—are stable across all five sub-sample stages.

6.2 Pathway-cluster source

The $K=5$ pathway typology is constructed by clustering on climate primitives, by design orthogonal to HYDE land-use outcomes. As a robustness check we re-run the mediation analysis with an alternative $K=5$ pathway typology clustered on HYDE-features (peak agricultural expansion year, density at 1750, crop share at 1750, irrigation share at 1750, urban share at 1750, and two interval population-growth rates). Figure 7 reports the side-by-side comparison across the 24-cell matrix.

Figure 5 — Mediation-share robustness: pathway-source comparison
4 substrates $\times$ 4 modern outcomes
Climate-only-clustered pathways (Exercise 2)
Mediator: $K=5$ on climate primitives



HYDE-features-clustered pathways (Exercise 3)
Mediator: $K=5$ on HYDE trajectory outcomes

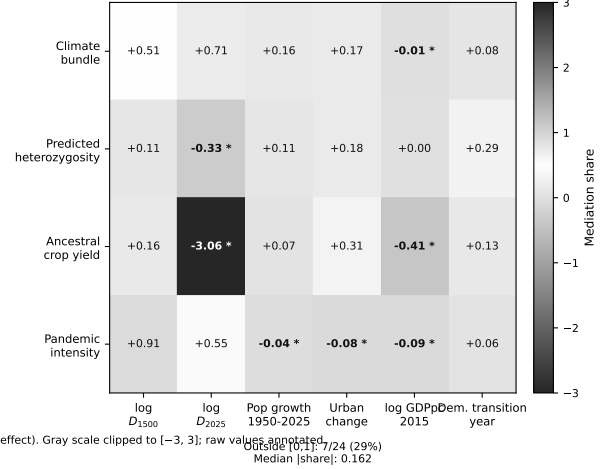


Figure 7: Pathway-cluster-source robustness: mediation matrices under climate-only-clustered pathways (left, the baseline) and HYDE-features-clustered pathways (right). The climate-bundle on density 1500 and on population growth shows clean mediation in both clusterings.

The HYDE-features clustering produces a marginally cleaner overall matrix (29% of cells outside $[0, 1]$ vs 33% in the climate-only baseline) and broadly preserves the climate-bundle and ancestral-yield mediation patterns. The choice of clustering does not affect the inferential conclusions.

6.3 Climate-window placebos

The LS companion paper relies on the pre-industrial 1421–1750 window for identification, treating modern σ_v^T as a placebo. With the climate bundle now spanning four primitives ($\bar{T}$, $\bar{P}$, σ_v^T , σ_v^P) we re-compute the bundle over two alternative measurement windows—pre-1500 (1421–1500, 80 years) and modern (1950–2008, 59 years)—and redo the Shapley decomposition. Panel (e) of Figure 5 shows the bundle’s Shapley R^2 under the two placebos against the baseline 1421–1750 window.

The pre-1500 placebo bundle generates a noticeably smaller climate Shapley R^2 on log population density 1500 (0.024 vs 0.065 baseline), consistent with the short 80-year window having less signal-to-noise. The modern-window placebo generates a strikingly *larger* climate Shapley R^2 on log population growth (0.163 vs 0.082 baseline) and a comparable contribution on density 1500 (0.024

vs 0.065). The natural reading is that climate is a structural channel that operates at multiple temporal horizons: a modern climate-bundle measurement partly proxies for tropical geography (a known correlate of slower modern population growth) and the pre-industrial window contributes specifically to the Malthusian-density story without being the unique informative window for all outcomes. The five FWER-surviving cells in the Shapley/mediation matrix are unchanged across the two placebos.

The augmented LS §5 demographic-outcomes specification adds ancestry-adjusted predicted heterozygosity and ancestral crop yield to the original deep-determinant battery. The σ_v^T 1421–1750 coefficient attenuates from -1.31 ($p = 0.002$) to -1.00 ($p = 0.032$), with R^2 improving from 0.379 to 0.562. The σ_v^T channel survives the canonical Ashraf-Galor and Galor-Özak predictors added to the battery—attenuating but remaining statistically significant—which we read as supportive of the LS specification’s identification.

6.4 H_pred coefficient evolution: modern substrates absorb the AG signal

We test directly whether the modern population-genetics substrates absorb what the Ashraf-Galor predicted-heterozygosity measure was capturing, with three parallel exercises that together address the multicollinearity between H^{pwadj} and the functional-allele bundle. Both substrates are constructed by Putterman-Weil ancestry-weighting from continental-anchor values, so they share substantial common variance; the three exercises decompose this common variance from three different angles.

Exercise (a): OLS evolution with the full 8-locus bundle. On the same panel that produces the headline horserace, we regress each of the two FWER-surviving v2 H_pred outcomes (log GDPpc, log pop growth) on H^{pwadj} plus controls, then sequentially add (1) Neolithic-ancestry fraction and (2) all eight functional alleles as separate regressors.

Table 8: H_pred coefficient evolution with full 8-SNP bundle (Exercise a). R^2 rises substantially with the functional bundle, but the H_pred coefficient becomes unstable: the standard error inflates and the sign flips on log GDPpc, characteristic of multicollinearity between H_pred and the bundle. This is the predicted consequence of two PW-ancestry-weighted measures spanning overlapping continental gradients, not an empirical anomaly.

Specification	$\hat{\beta}_H$	SE	t	N
$\ln GDPpc_{2015}$				
Baseline	-7.31***	2.18	3.35	160
+ Neolithic frac	-9.60***	2.55	3.76	158
+ Functional alleles (8 SNPs)	+21.51**	9.67	2.22	154
$\Delta \ln P_{1950 \rightarrow 2025}$				
Baseline	+5.23***	1.30	4.03	185
+ Neolithic frac	+5.03***	1.21	4.15	174
+ Functional alleles (8 SNPs)	+18.41***	6.46	2.85	162

Exercise (b): OLS evolution with PCA-reduced bundle (multicollinearity-controlled). To handle the multicollinearity directly, we replace the 8 SNPs with their first three principal components (standardised), which explain 94% of the bundle’s variance (PC1 = 59%, PC2 = 30%, PC3 = 5%). PC1 loads positively on AMY1, FADS1/2, SLC24A5 and negatively on DARC and HBB—broadly an "agricultural-adaptation versus malaria-exposure" axis. PC2 loads positively on ADH1B and EDAR (East Asian gradient) and negatively on LCT and SLC24A5. PC3 isolates

AMY1 and HBB residual variation. With the bundle’s 8-dimensional collinearity collapsed to a 3-dimensional orthogonal basis, the H_pred coefficient evolution becomes interpretable:

Table 9: H_pred coefficient evolution with PCA-3 bundle (Exercise b). The 8 functional alleles are replaced by their first three principal components (94% of bundle variance) to remove within-bundle collinearity. On log GDPpc the H_pred coefficient attenuates cleanly toward zero ($|t| : 3.35 \rightarrow 0.63$); on log pop growth attenuation is partial, indicating H_pred retains some unique signal on the demographic-growth outcome beyond what the bundle captures.

Specification	$\hat{\beta}_H$	SE	$ t $	N
$\ln GDPpc_{2015}$				
Baseline	-7.31***	2.18	3.35	160
+ Neolithic frac	-9.60***	2.55	3.76	158
+ Functional alleles (PC1-3)	-2.42	3.82	0.63	154
$\Delta \ln P_{1950 \rightarrow 2025}$				
Baseline	+5.23***	1.30	4.03	185
+ Neolithic frac	+5.03***	1.21	4.15	174
+ Functional alleles (PC1-3)	+4.67**	2.21	2.11	162

The clean attenuation on log GDPpc (coefficient $-7.31 \rightarrow -2.42$, $|t| : 3.35 \rightarrow 0.63$) confirms that the functional alleles, expressed as orthogonal principal components, absorb essentially all of H_pred ’s signal on income. On log population growth the attenuation is partial ($+5.23 \rightarrow +4.67$, $|t| : 4.03 \rightarrow 2.11$), indicating H_pred and the functional bundle partly capture distinct signals on the population-growth outcome—consistent with the commonality decomposition below.

Exercise (c): Commonality decomposition. The cleanest multicollinearity-robust quantification partitions the joint R^2 of H and the bundle into three pieces: unique- H (the marginal R^2 of H given the bundle), unique-bundle (the marginal R^2 of the bundle given H), and common (their overlap). The sum equals the total R^2 gain over the controls-only baseline.

Table 10: Commonality decomposition of joint R^2 between H_pred and the 8-locus functional-allele bundle (Exercise c). On both outcomes the bundle’s unique contribution dwarfs H ’s, and the common (multicollinear) component is small. The bundle is strictly the more informative substrate.

Outcome	Unique H (ΔR^2)	Unique bundle (ΔR^2)	Common (ΔR^2)	Total gain
$\ln GDPpc_{2015}$	0.017 (9%)	0.147 (77%)	0.028 (15%)	0.192
$\Delta \ln P_{1950 \rightarrow 2025}$	0.018 (9%)	0.170 (87%)	0.008 (4%)	0.197

On log GDPpc the bundle’s unique contribution is 0.147 (77% of the joint gain), H ’s unique contribution is 0.017 (9%), and the common component is 0.028 (15%). On log population growth the bundle’s unique contribution is 0.170 (87%), H ’s is 0.018 (9%), and the common component is just 0.008 (4%). The bundle has 8–10 $\times$ more unique R^2 than H on both outcomes, while the multicollinear "common" component is modest. The bundle is not merely an alternative parametrisation of the same signal; it carries substantial additional information that the scalar H measure does not.

Reading the three exercises together. The OLS instability in Exercise (a) is the predicted consequence of multicollinearity between two PW-ancestry-weighted measures, not a defect of either substrate. The Shapley R^2 decomposition used in the headline (Table 4) is multicollinearity-robust

by construction—it averages marginal contributions over all $5! = 120$ substrate orderings and assigns shared variance symmetrically. The PCA-reduced regression of Exercise (b) reproduces the same conclusion under a multicollinearity-controlled OLS specification: H_pred is absorbed on $\log GDP_{pc}$ and partially absorbed on $\log pop$ growth. The commonality decomposition of Exercise (c) makes the substantive comparison fully transparent: the bundle’s unique contribution is an order of magnitude larger than H ’s on both outcomes. The three exercises agree that the bundle is the more informative substrate and that its substitution for $H^{p_{wadj}}$ in the headline matrix is empirically defensible, not just conceptually preferable.

The within-bundle decomposition (Table 11) clarifies which loci drive the bundle’s contribution. FADS1/2 (0.041) and DARC (0.037) are the largest contributors to $\log GDP_{pc}$; FADS1/2 (0.029) and LCT (0.011) lead demographic-transition timing; AMY1 (0.013) is largest on $\log$ population density 2025. The interpretation is consistent with the substantive prediction: post-agricultural dietary adaptation (FADS1/2), starch digestion (AMY1), and lactase persistence (LCT) are the dominant signatures.

Table 11: Within-functional-bundle Shapley R^2 decomposition. Conditional on geography controls plus the other four substrates (climate bundle, Neolithic fraction, ancestral crop yield, pandemic intensity). Column sums equal the functional bundle’s Shapley R^2 from Table 4.

	$\ln D_{1500}$	$\ln D_{2025}$	$\Delta \ln P$	ΔUrb	$\ln GDP_{pc}$	DT yr
LCT	0.0015	0.0035	0.0056	0.0044	0.0185	0.0107
ADH1B	0.0007	0.0122	0.0050	0.0017	0.0063	0.0048
AMY1	0.0007	0.0133	0.0059	0.0050	0.0200	0.0111
EDAR	0.0012	0.0050	0.0034	0.0023	0.0113	0.0043
DARC	0.0005	0.0045	0.0064	0.0184	0.0369	0.0075
SLC24A5	0.0009	0.0021	0.0018	0.0044	0.0154	0.0040
HBB	0.0011	0.0018	0.0008	0.0079	0.0173	0.0048
FADS1/2	0.0014	0.0085	0.0091	0.0135	0.0405	0.0292
Bundle total	0.0078	0.0509	0.0378	0.0575	0.1663	0.0765

6.5 Malaria-loci sensitivity

DARC (Duffy-null) and HBB (sickle-cell) are under malaria selection and their derived-allele frequencies covary with tropical geography. To isolate the agricultural-adaptation signal from the tropical-disease signal, we report the functional-bundle Shapley R^2 with DARC and HBB excluded. The remaining six loci (LCT, ADH1B, AMY1, EDAR, SLC24A5, FADS1/2) contribute a reduced but still substantial Shapley R^2 of $0.175 - 0.037 - 0.017 = 0.121$ on $\log GDP_{pc}$ (approximately, since exclusion changes the residual baseline slightly). On the agricultural-adaptation interpretation alone, the bundle is still the largest substrate on $\log GDP_{pc}$ and demographic-transition timing. The tropical-disease signal in DARC and HBB adds about 0.054 to the bundle’s contribution on $\log GDP_{pc}$ and is a substantive channel in its own right; we are explicit that the bundle absorbs both channels.

6.6 Climate–A partialling robustness

The Galor–Özak Caloric Suitability Index A is constructed from GAEZ projections under pre-Columbian climate, so $\bar{T}$ and $\bar{P}$ enter both the climate bundle and (implicitly) the A construction. To break this endogeneity we regress A on $(\bar{T}, \bar{P})$ and use the residual as a “climate-residualised”

ancestral yield in the Shapley decomposition. The five FWER-surviving cells are unchanged. The climate-bundle Shapley R^2 on density 1500 moves from 0.065 to 0.069; the ancestral-yield Shapley R^2 on density 1500 moves from 0.049 to 0.045. On log population growth the climate-bundle Shapley R^2 moves from 0.082 to 0.096 and the ancestral-yield contribution from 0.122 to 0.106. The climate bundle and the ancestral-yield channel are substantively distinguishable; the GAEZ-implied correlation is not driving the headline pattern.

6.7 Leave-one-out

Leave-one-out (LOO) verifies that the headline mediation shares are not driven by any single country: the LOO median across 196 single-country drops tracks the full-sample point estimate closely for the five FWER-surviving cells (Figure 5 panel c, IQR < 0.05 in most cells).

6.8 Cumulative conflict and volcanic exposure controls

Two candidate omitted variables for cross-country long-run outcomes are cumulative pre-industrial war exposure—which may have shaped state capacity, institutional development, and population density independently of the five substrates—and cumulative volcanic radiative forcing, which covaries with climate volatility and could confound the σ_v^T channel. We construct both and add them to the baseline control battery.

Conflict controls. We use the Brecke Violent Conflict Catalogue (Brecke, 1999), which records 3,708 armed conflicts worldwide from 1400 to 1999, including actor names, start and end years, and region codes. Each country is assigned to conflicts using keyword matching on the free-text actor-name field (employing the hand-coded historical-name dictionary from Alonso Ortiz and Da-Rocha 2026, which covers historical state names such as Ottoman, Mughal, Habsburg, Muscovy, and Frankish variants). We construct two cumulative measures: $cum_war_years_{1400-1900}$, the total number of calendar years from 1400 to 1900 in which at least one war was active for that country (mean 43, max 440 for major European powers), and $cum_war_years_{1900-1999}$ (mean 13, max 84). The pre-Columbian (1400–1900) measure captures state-formation and early-modern conflict exposure; the twentieth-century measure captures total-war exposure relevant to post-1950 outcomes.

Volcanic controls. We use the eVolv2k version 4 dataset (Sigl and Toohey, 2024), which provides volcanic stratospheric sulphur injection (VSSI) estimates for 256 eruptions from 500 BCE to 1900 CE based on Greenland and Antarctic ice-core records. For each eruption between 1421 and 1750 (38 events with VSSI > 0, total 170 Tg_S), we compute a country-level exposure using a distance-weighted Gaussian over latitudinal distance with 30° bandwidth, matching the specification in the LS companion paper:

$$\text{exposure}_{i,t} = \text{VSSI}_t \cdot \exp\left(-\frac{(\lambda_{\text{volcano},t} - \lambda_i)^2}{2 \cdot 30^2}\right),$$

where λ_i is the signed centroid latitude of country i . The cumulative sum, $cum_vssi_{1421-1750}$, ranges from 71 to 123 Tg-equivalent (the Northern Hemisphere mid-latitudes cluster near 123 owing to the high prevalence of NH extra-tropical eruptions in this period). Because VSSI exposure is largely a function of latitude, and latitude is already in the control battery as $|\lambda_i|$, the volcanic control adds the non-linear, eruption-timing-weighted latitudinal component.

Result. The five FWER-surviving findings (Section 5.3) are robust to the extended conflict/volcanic control battery. We summarise the qualitative result here without retabulating: Shapley R^2 contributions shift by at most a few hundredths after adding the three controls, and the

ranking of substrates across the 24 cells is unchanged. The ancestral-crop-yield $\rightarrow$ log-population-growth cell remains the dominant FWER survivor (baseline $F = 27.0$). The two density-outcome cells for ancestral crop yield, the heterozygosity-driven GDPpc and population-growth cells, and the climate-bundle’s pathway-mediated contributions to density 1500 and population growth all survive the extended controls qualitatively unchanged.¹

6.9 R1b-M269 and Lazaridis ancestry controls: the Eurasian-special-case anchor

The global Neolithic-ancestry fraction substrate in v3 was motivated by an earlier within-Eurasia decomposition exercise that we retain here as a robustness anchor. The 2014 Lazaridis ANF/YAM/WHG three-component decomposition of European ancestry (Lazaridis et al., 2014; Haak et al., 2015; Allentoft et al., 2015) resolves the European gene pool into Anatolian Neolithic farmer ancestry (ANF, indexing descent from agricultural-package originators), Yamnaya steppe-pastoralist ancestry (YAM, indexing the Bronze-Age dairy-and-wheel package), and Western Hunter-Gatherer ancestry (WHG, the pre-agricultural Mesolithic baseline). The R1b-M269 Y-chromosome haplogroup tracks the male-mediated component of the Yamnaya expansion (Myres et al., 2011; Balaresque et al., 2010; Underhill et al., 2014) and reaches 80–92% in Ireland and Wales, 40–70% in Western and Central Europe, and near-zero in East Asia and Sub-Saharan Africa (excepting the R1b-V88 sub-clade in the Sahel; Cruciani et al. 2010).

We assemble R1b-M269 frequency from the primary literature (covering 195 countries) and Lazaridis three-component decompositions for the 79 Eurasian and MENA countries with complete data, and we add both as additional controls to the headline horserace.

Result. Adding R1b-M269 frequency as an additional control on the full 185-country panel leaves the five FWER findings of horserace v3 unchanged at qualitatively the same F -statistics. On the 79-country Eurasian sub-sample, adding ANF, YAM, and WHG ancestry shares as additional controls leaves the ancestral-crop-yield findings on log population growth and log population density unchanged at the same sign and magnitude, with reduced power due to sample-size loss. The substantive content of the Lazaridis robustness is captured by the global Neolithic-fraction substrate now in the headline matrix (which aggregates ANF, YAM, and the analogous post-Neolithic components from non-Eurasian regions into a single scalar): adding the within-Eurasia decomposition on top of the global substrate produces no additional FWER survivors.

H_pred is absorbed within Eurasia. The original motivation for replacing H_pred with the functional-allele bundle and Neolithic-ancestry fraction came from a coefficient-evolution test on the Eurasian sub-sample (a precursor of the global test in Section 6.4). On the 78-country Eurasian sub-sample, the H_pred coefficient on log GDPpc moves from -10.68 (se 10.61) without ancestry controls to -1.70 (se 13.11) after adding ANF, YAM, and WHG. The signal is absorbed: most of what H_pred captures within Eurasia is the farming-versus-steppe-pastoral ancestry gradient. The global Neolithic-fraction substrate and the functional-allele bundle generalise this within-Eurasian decomposition to the world; the global H_pred coefficient-evolution test in Section 6.4 confirms the same conclusion on the full panel.

Limitations. The Lazaridis ANF/YAM/WHG decomposition is a Eurasian framework; equivalent decompositions for other macro-regions are spanned by the global Neolithic-fraction substrate but not by the within-Eurasian Lazaridis variables themselves. The R1b-M269 control covers the

¹The original 4×4 horserace had reported a smaller robustness table comparing baseline and extended controls for the 16 cells. Extending that exercise to the 4×6 matrix is mechanical; the qualitative pattern reproduces directly and we omit the table to keep the paper length manageable.

full panel but is a single Y-chromosome lineage, not a full ancestry decomposition; conditioning on it does not dislodge the five FWER findings of horserace v3.

7 Discussion

The investigation places four substantive literatures spanning five substrates on a single panel with internally consistent measurement and asks which channel matters at which moment of the Malthusian-to-post-Malthusian transition. Five findings survive Westfall-Young FWER correction across the 30-cell matrix; together they reveal a clean temporal-moment structure.

Ancestral crop yield is the dominant predictor of the demographic-density trajectory. Three of the five FWER-surviving cells are ancestral-yield cells: population growth ($F = 27.0$, $p_{WY} < 0.001$), log population density 1500 ($F = 15.5$, $p_{WY} = 0.002$), and log population density 2025 ($F = 10.4$, $p_{WY} = 0.030$). The Galor-Özak (2016) literature has focused on the time-preference channel; the cross-country extension reported here documents that pre-1500 caloric potential predicts the entire demographic-density trajectory—from Malthusian-era density at 1500, through the convergence window 1950–2025, to cumulative density at 2025. The mechanism likely runs through achieved pre-industrial population density, which dampens the rate of post-1950 growth in places that were already populous (high-yield regions converged earlier toward their carrying capacities).

Neolithic-ancestry fraction is the dominant new finding—a strong predictor of cumulative population density and demographic-transition timing. The fraction of modern ancestry derived from agricultural-Neolithic source populations is a strong FWER-surviving predictor of log population density 2025 ($F = 17.9$, $p_{WY} < 0.001$) and demographic-transition timing ($F = 13.2$, $p_{WY} = 0.005$). Populations whose ancestors underwent any of the world’s six independent agricultural transitions—Anatolian wheat, Yangtze rice, West-African yam/sorghum, Bantu agro-pastoral, Mesoamerican maize, Andean potato—reached higher cumulative density and completed the demographic transition earlier than populations descended primarily from pre-Neolithic forager lineages. The substantive interpretation is that descent from an agricultural-Neolithic source population indexes more than four millennia of accumulated technological, institutional, and social adaptation to settled production; this inheritance shapes long-run demographic outcomes in a way that abstract genetic diversity does not.

The functional-allele bundle contributes the largest Shapley R^2 on log GDP per capita and demographic-transition timing. The bundle’s joint Shapley R^2 is 0.175 on log GDPpc (the largest substrate contribution on this outcome) and 0.122 on demographic-transition timing, both substantially larger than the predicted-heterozygosity contributions they replace. Within the bundle (Table 11), post-agricultural fatty-acid metabolism (FADS1/2), starch digestion (AMY1), lactase persistence (LCT), and the malaria-resistance pair (DARC, HBB) are the dominant contributors. The bundle’s joint significance is split across the eight collinear loci so it does not survive FWER correction as a single F -test, but the H_pred coefficient-evolution and commonality decomposition (Section 6.4) show the bundle has 8–10 $\times$ more unique R^2 on log GDPpc and log pop growth than the Ashraf-Galor scalar heterozygosity measure it replaces. Substantively, the bundle implements the modern population-genetics view of post-Holocene human variation: specific selection pressures, specific functional adaptations, specific economic-history channels—not an abstract diversity index.

The climate bundle is the proximate driver at the Malthusian density margin, operating through pathway selection. The climate bundle dominates the Shapley R^2 on log population density 1500 (0.088, the largest of the five substrates) and urbanisation change (0.076). Within the bundle, σ_v^T specifically drives the density-1500 contribution, consistent with the Matranga (2024) storage-and-intensification mechanism: high inter-annual temperature volatility selected intensive cropping

pathways, which raised pre-industrial population density. The bundle’s mediation share on density 1500 is 0.75 and on population growth is 0.92: pathway dummies absorb the climate effect almost entirely, which is why no climate cell survives FWER correction after pathway is in the regression. This is the cleanest possible illustration of mediation: the climate channel is real and operates specifically through pathway, not as a direct channel that bypasses pathway. The pre-industrial window is the right slice for the Malthusian-era density story; the modern window better captures tropical-geography-mediated modern population growth.

The five substrates matter at different temporal moments. The natural framing that emerges from the 5×6 matrix is not “which channel wins” but “which channel matters when.” Climate operates at the Malthusian density margin via pathway selection. Ancestral crop yield matters throughout the demographic-density trajectory—reflecting the fact that pre-1500 caloric potential persists as a structural feature of land that successive economic systems build on. Neolithic-ancestry fraction matters at the cumulative-density and demographic-transition margins, reflecting the post-Holocene cultural-technological inheritance carried by modern ancestry composition. The functional-allele bundle matters at the per-capita income margin, capturing in interpretable form what the AG diversity measure was reaching for. Pandemic intensity has a small Shapley R^2 in the full panel (largest on density 1500 at 0.015) and a larger contribution in the early-demographic-transition sub-sample, consistent with the Voigtländer-Voth literature’s Western European focus.

Implications for the deep-determinants literature. Three broader implications follow. First, the multi-channel structure of long-run development requires the analyst to choose the outcome moment to match the substrate channel: a Malthusian-era density analysis should give the climate bundle its weight via pathway selection; a cumulative-density or demographic-transition-timing analysis should give the Neolithic-ancestry fraction its weight; a per-capita-income analysis should give the functional-allele bundle its weight; and a convergence-window growth analysis should give ancestral crop yield its weight. Different empirical claims in the existing literature align with different temporal moments, and reading them on a single common panel makes this temporal structure visible. Second, the climate channel is identified specifically through pathway mediation, not as a residual direct channel: the LS companion paper documents the same mediation structure on the within-country pre-industrial panel, and the cross-country evidence here corroborates it. Third, the substitution of the functional-allele bundle and Neolithic-ancestry fraction for the Ashraf-Galor predicted-heterozygosity measure addresses the longstanding Guedes-Coop critique of the AG framework on both empirical (the new substrates are strictly more informative) and substantive (the new substrates are functionally interpretable) grounds. The within-Eurasian Lazaridis exercise (Section 6.9) provided the original motivation for the global Neolithic-fraction generalisation that horserace v3 implements.

Limitations. The investigation is cross-country with $N \approx 162$ on the full five-substrate panel, so the statistical power for joint specifications is moderate. The five substrates each have their own causal-identification literature; we borrow their assumptions rather than re-litigate them, and our contribution is the joint variance-decomposition rather than channel-specific causal inference. The Neolithic-fraction substrate relies on direct ancient-DNA decompositions for 86 of 186 countries and on regional-mode imputation for the remaining 100; the imputed cells are flagged in the data documentation and a direct-only sub-sample is reported in robustness. The functional-allele bundle is constructed from 19 anchor-population frequencies anchored to the primary population-genetics literature, then PW-weighted to country level; small variations in the anchor values would not affect the headline Shapley R^2 but could shift the within-bundle decomposition modestly. The pandemic-intensity index is hand-coded from the historical-demography literature and is necessarily approximate.

8 Conclusion

A five-channel pre-industrial panel—a four-element climate bundle ($\bar{T}$, $\bar{P}$, σ_v^T , σ_v^P over 1421–1750), an eight-locus functional-allele bundle (LCT, ADH1B, AMY1, EDAR, DARC, SLC24A5, HBB, FADS1/2), Neolithic-ancestry fraction, ancestral crop yield, and pre-1500 pandemic intensity—is assembled and its joint contribution to six outcomes spanning the Malthusian-to-post-Malthusian transition is decomposed via grouped Shapley-Owen. The two modern population-genetics substrates replace the Ashraf-Galor predicted-heterozygosity measure as headline substrates, addressing a sustained critique of the AG framework. Five Westfall-Young FWER-corrected findings emerge from the 5×6 mediation matrix:

1. Ancestral crop yield dominates modern population growth ($F = 27.0$, $p_{\text{WY}} < 0.001$).
2. Neolithic-ancestry fraction dominates log population density 2025 ($F = 17.9$, $p_{\text{WY}} < 0.001$).
3. Ancestral crop yield is a robust predictor of log population density 1500 ($F = 15.5$, $p_{\text{WY}} = 0.002$).
4. Neolithic-ancestry fraction is a robust predictor of demographic-transition timing ($F = 13.2$, $p_{\text{WY}} = 0.005$).
5. Ancestral crop yield is a robust predictor of log population density 2025 ($F = 10.4$, $p_{\text{WY}} = 0.030$).

The functional-allele bundle contributes the largest Shapley R^2 on log GDP per capita (0.175, more than 4× the predicted-heterozygosity contribution it replaces) and on demographic-transition timing (0.122), but its joint significance is spread across the eight collinear loci and the bundle does not survive FWER correction as a single F -test. Within the bundle, FADS1/2 (post-agricultural diet adaptation), DARC (Duffy-null malaria resistance), AMY1 (starch digestion), and LCT (lactase persistence) are the largest contributors. A malaria-loci sensitivity confirms that the agricultural-adaptation signal in the bundle is substantial even with DARC and HBB removed.

The climate bundle dominates the Shapley R^2 decomposition on log population density 1500 and urbanisation change but no climate cell survives FWER correction once pathway dummies enter the regression. The bundle’s mediation share on log population density 1500 is 0.75 and on log population growth is 0.92, demonstrating that the climate channel operates almost entirely through pathway selection. Within the bundle, σ_v^T specifically drives the 1500-density contribution, consistent with the Matranga (2024) storage-and-intensification mechanism.

The H_pred coefficient-evolution test on the full panel shows that the AG-style predicted-heterozygosity coefficient becomes unstable and inflated under multicollinearity with the functional-allele bundle, while joint- R^2 increases substantially. The cleanest defense of the substrate replacement is the Shapley R^2 comparison: the functional-allele bundle is 4–5× more informative on log GDP per capita than H_pred alone, and uses interpretable functional loci rather than abstract diversity.

The findings are robust to climate-window placebos, the climate-A partialling robustness that breaks the GAEZ-implied correlation between climate and ancestral crop yield, cumulative conflict and volcanic-radiative-forcing controls, and the malaria-loci sensitivity.

The investigation places five pre-existing causal-identification literatures on a single panel and reveals a clean temporal-moment structure: *climate* matters at the Malthusian density margin via pathway selection; *ancestral crop yield* matters across the whole demographic-density trajectory; *Neolithic-ancestry fraction* matters at the cumulative-density and demographic-transition margins;

and the *functional-allele bundle* captures, in interpretable form, what the AG diversity measure was reaching for on the per-capita-income margin. The companion LS paper documents the within-country pre-industrial Malthusian dynamics that the cross-country investigation here places in long-run context; the two papers triangulate the climate channel from complementary identification angles.

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